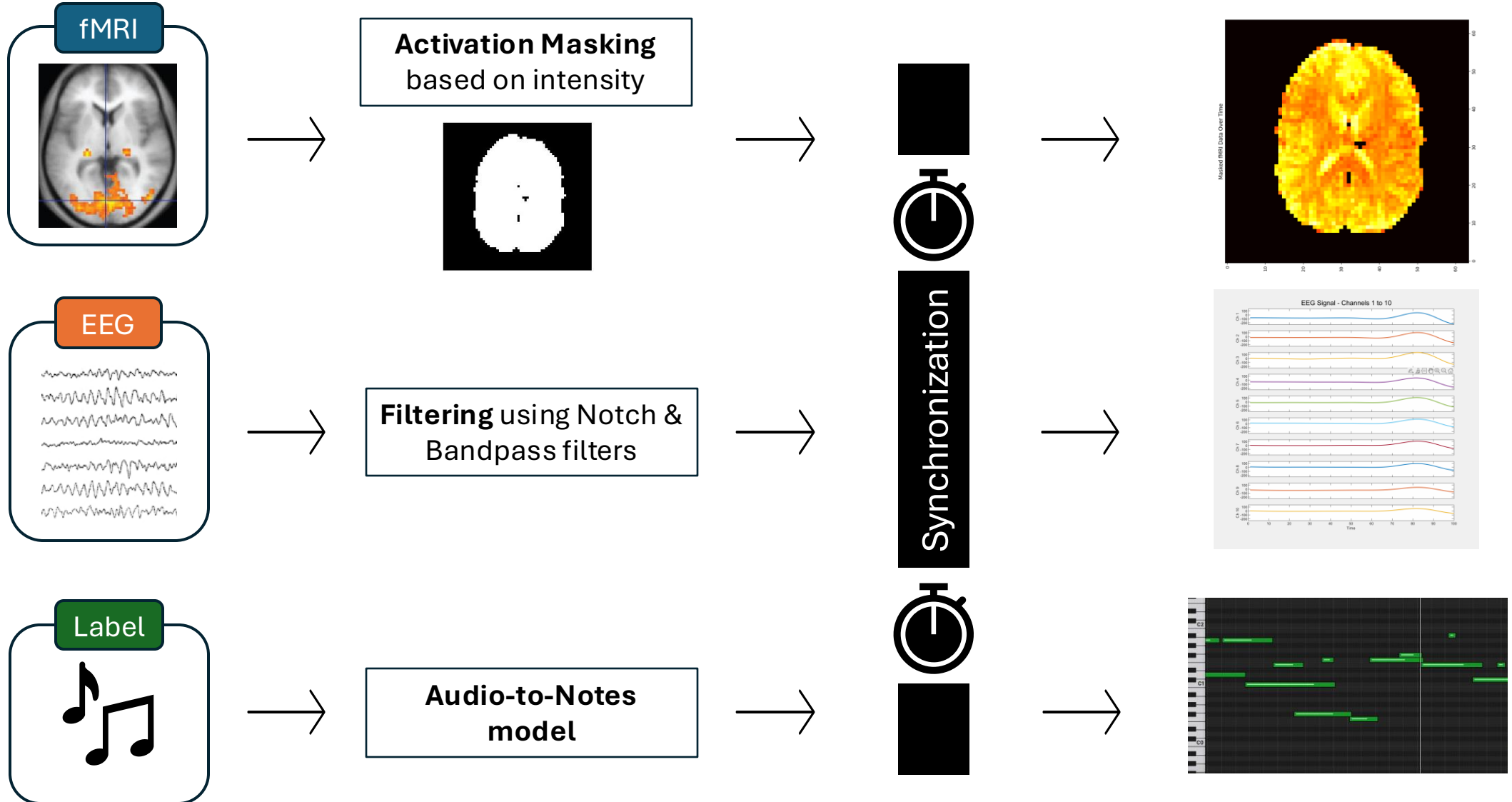


MIND-FUSE

Multimodal Interpretability-Driven Fusion and
Embedding

Pre-Processing of Data



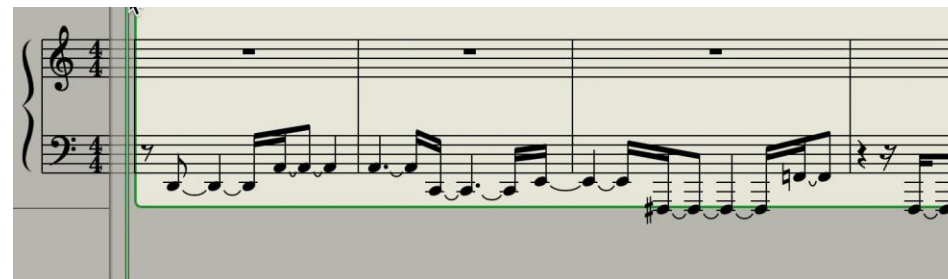
Audio to Features Pipeline

Original
Generated
Track

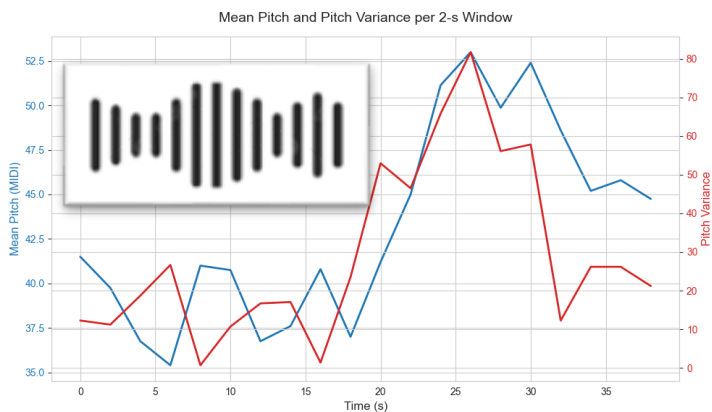


Audio-to-Midi Model

Transformer-based
model (Transkun)
From Yan, 2024*

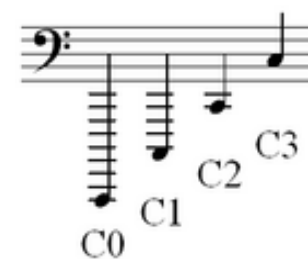


Midi to Notes Translation



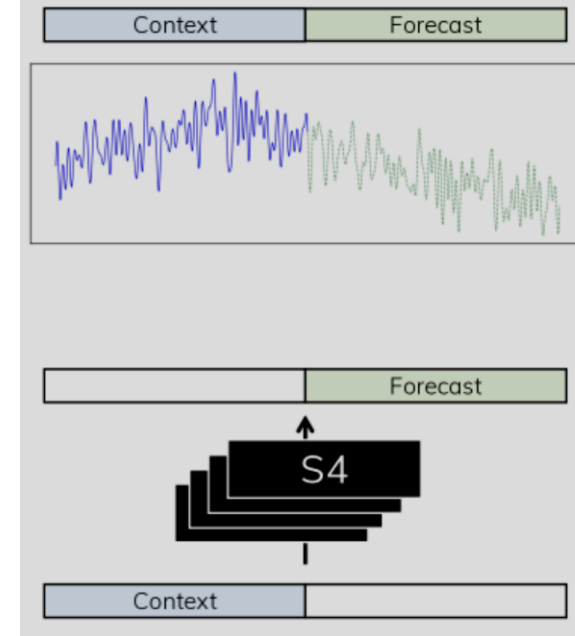
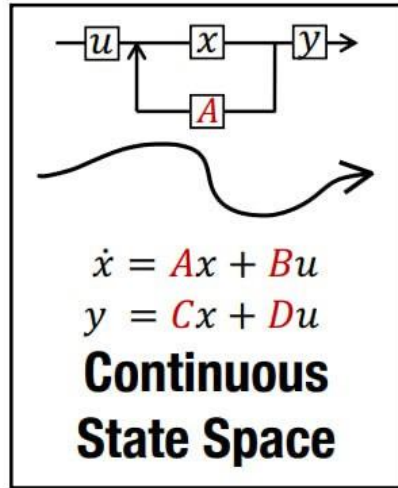
Average Notes

Variance in Notes



[24, 32, 36, 48]

Temporal Encoder S₄



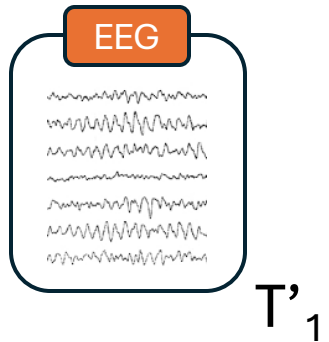
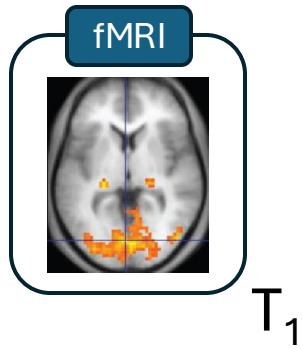
- Encodes temporal information up to n time points
- Reduces information loss normally seen in downsampling
- Facilitates temporal alignment of neuroimaging signals with arbitrary frequencies

[Submitted on 31 Oct 2021 (v1), last revised 5 Aug 2022 (this version, v3)]

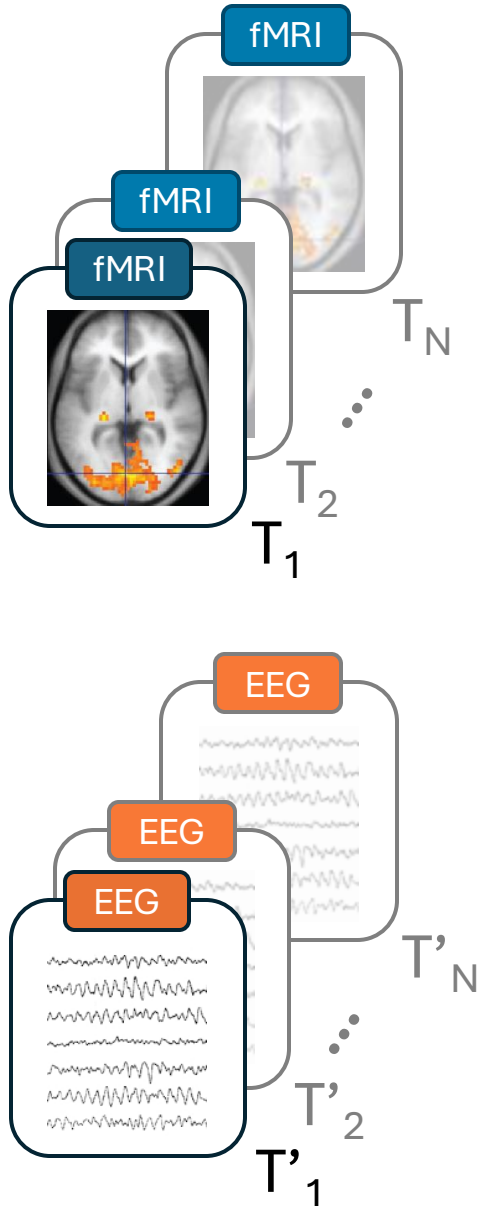
Efficiently Modeling Long Sequences with Structured State Spaces

Albert Gu, Karan Goel, Christopher Ré

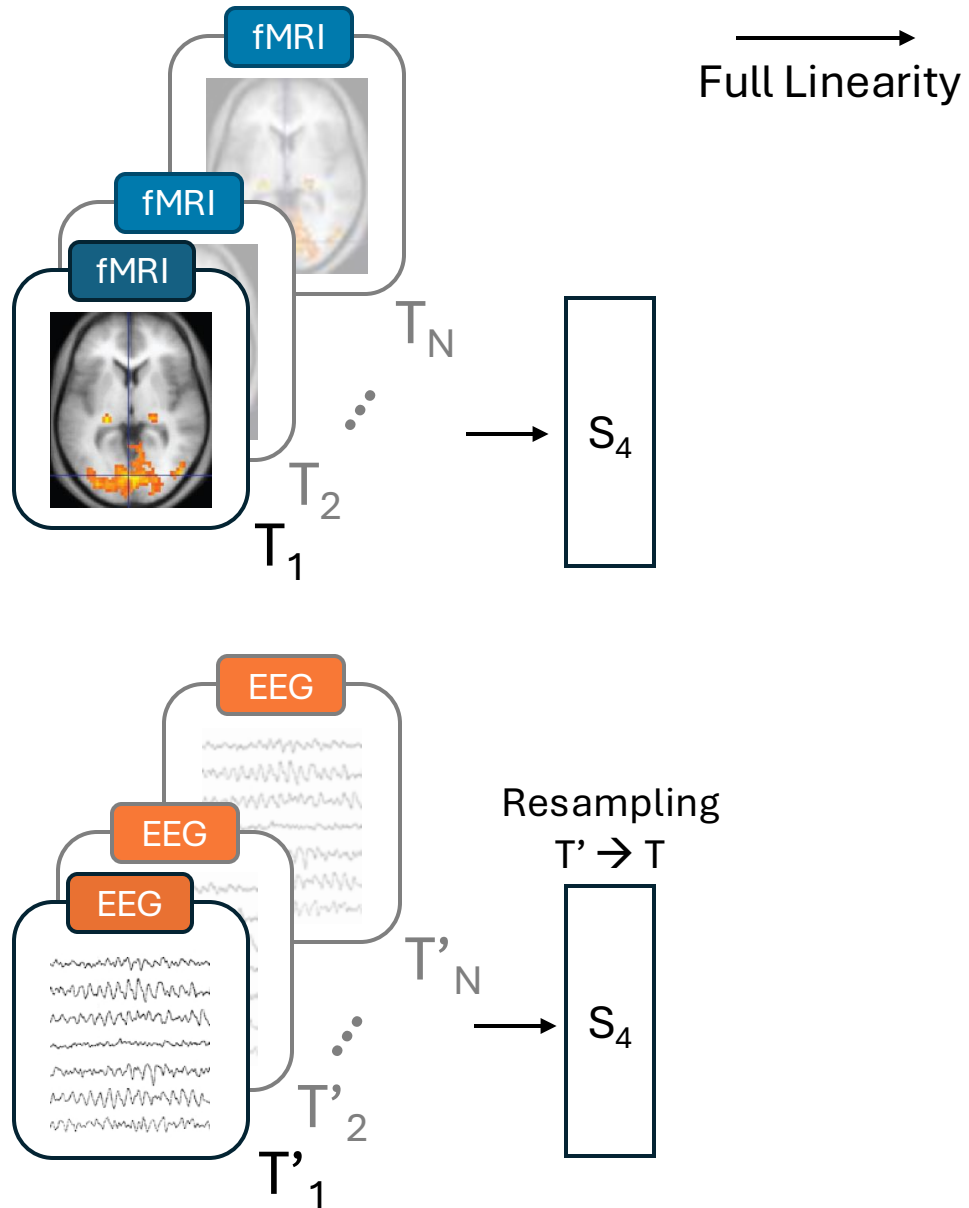
Supervised Cross-Modal Fusion



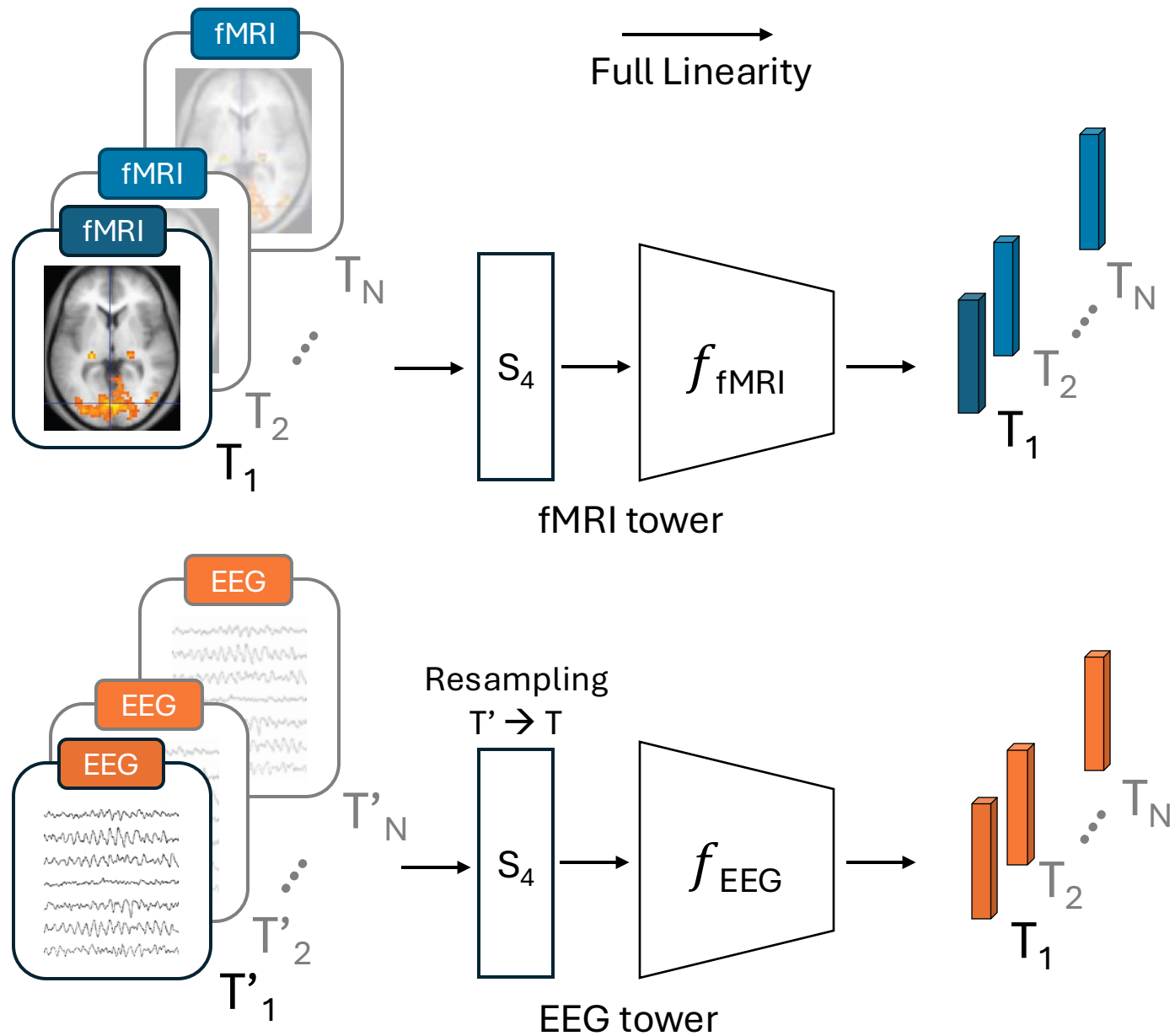
Supervised Cross-Modal Fusion



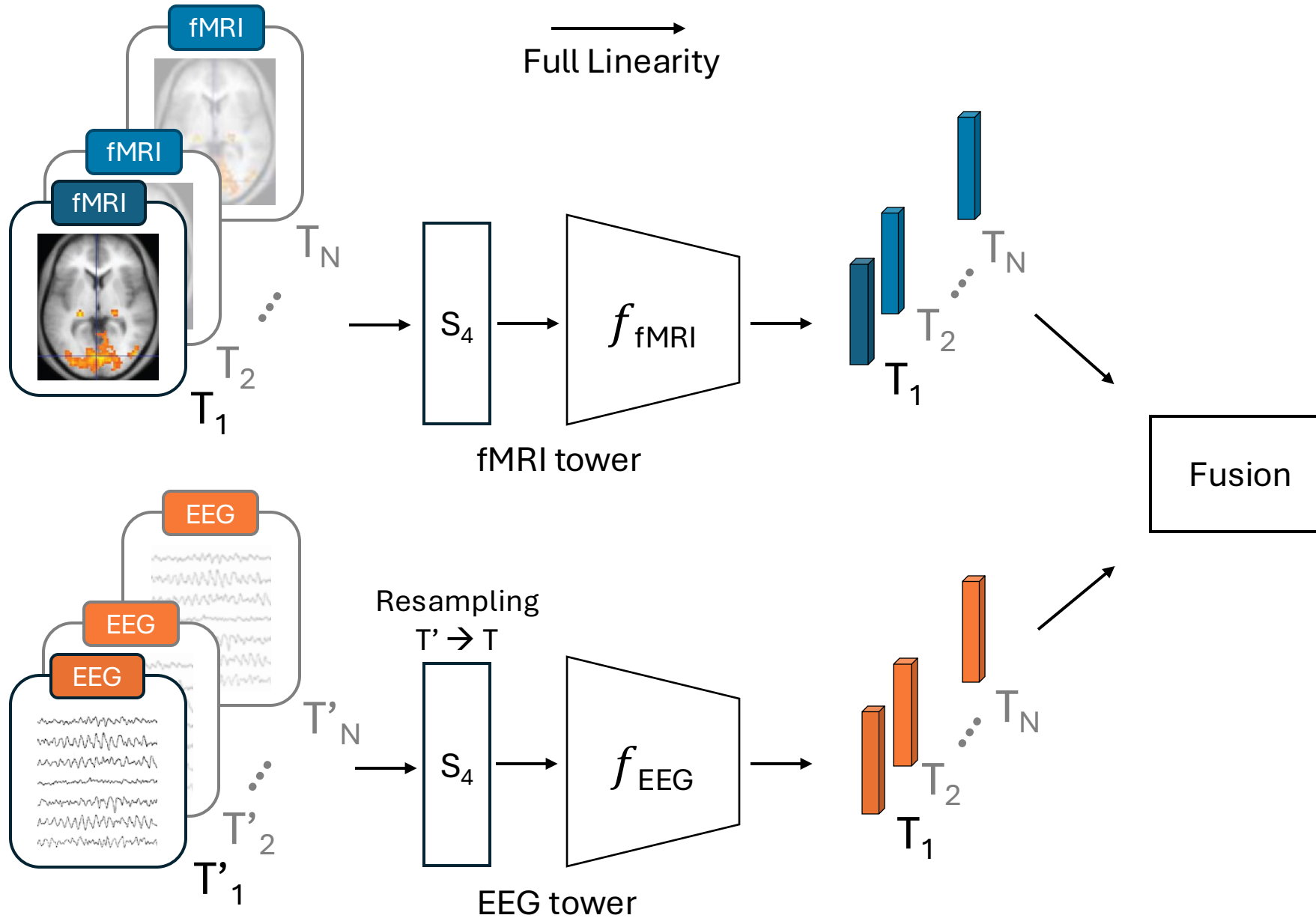
Supervised Cross-Modal Fusion



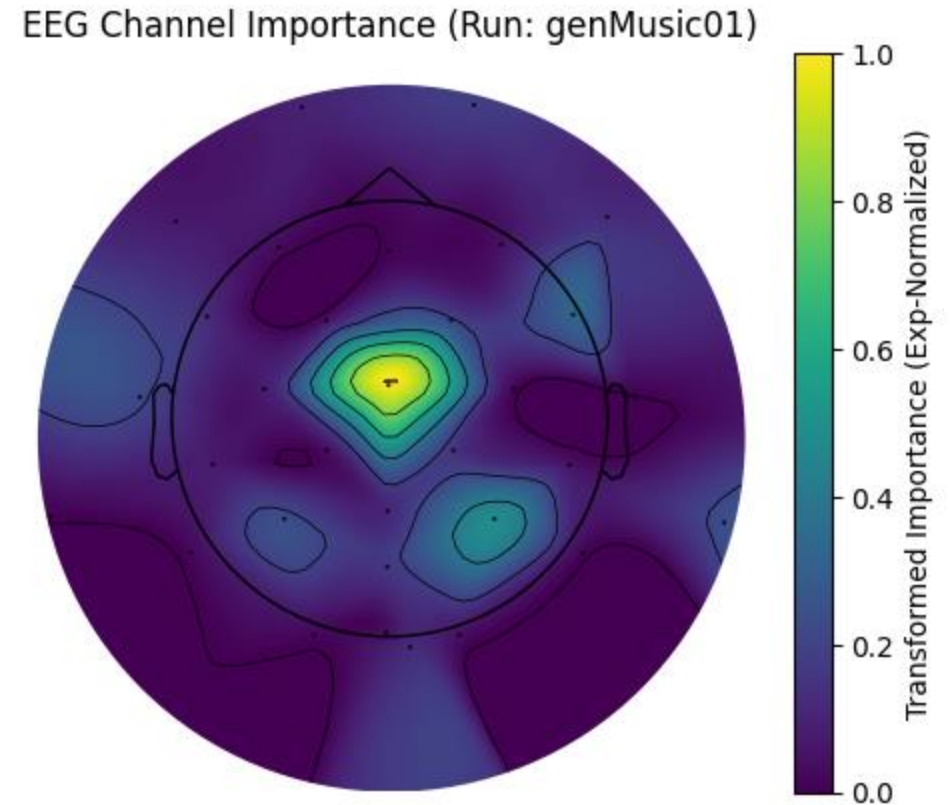
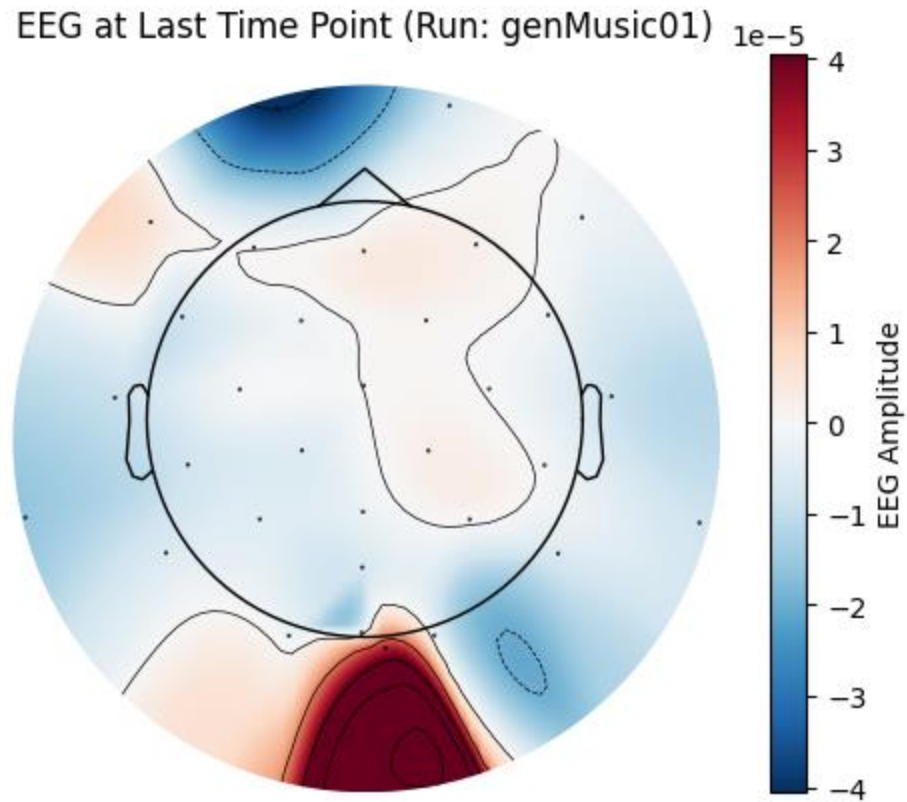
Supervised Cross-Modal Fusion



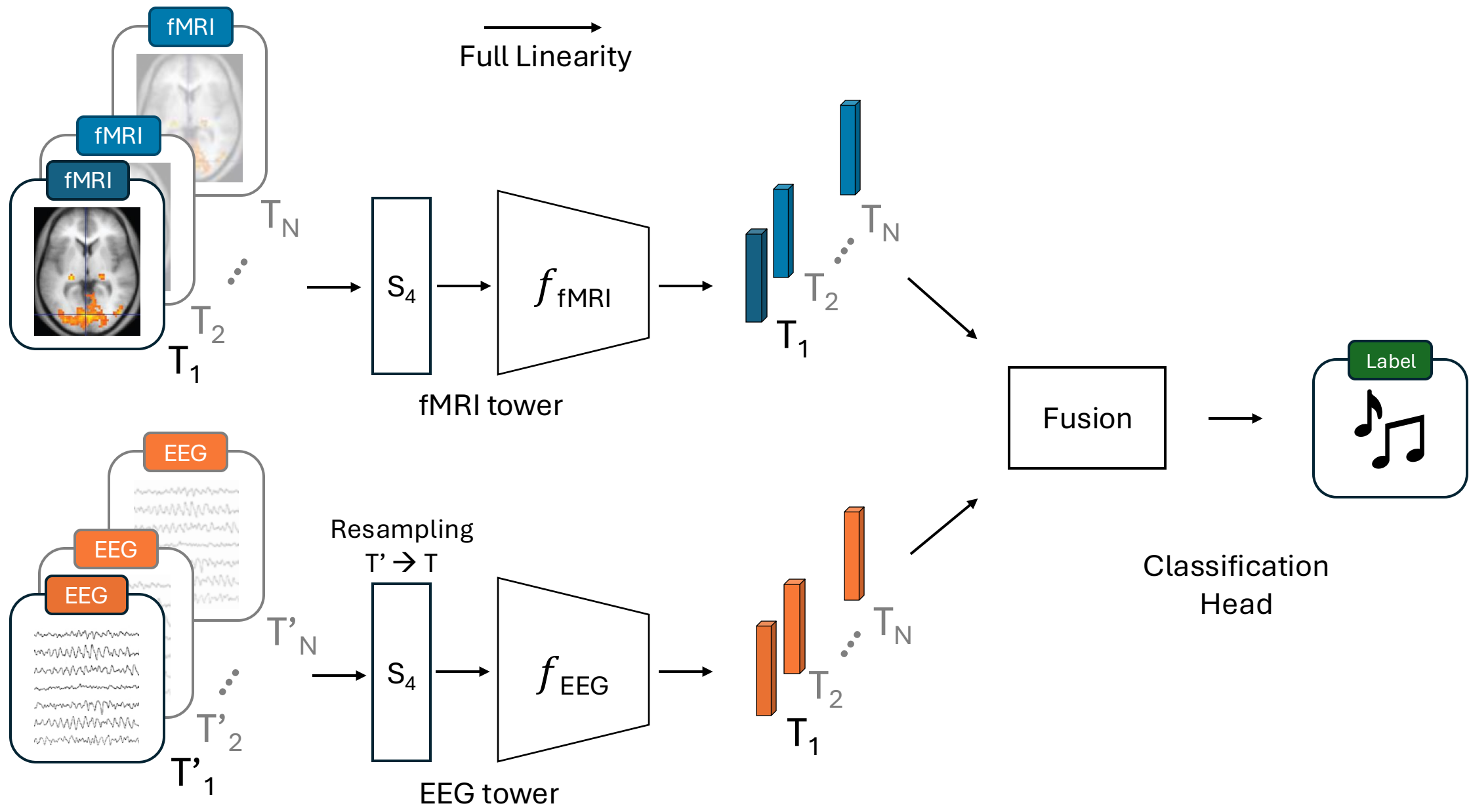
Supervised Cross-Modal Fusion



Linearity Allows Feature Interpretability of Joint Space

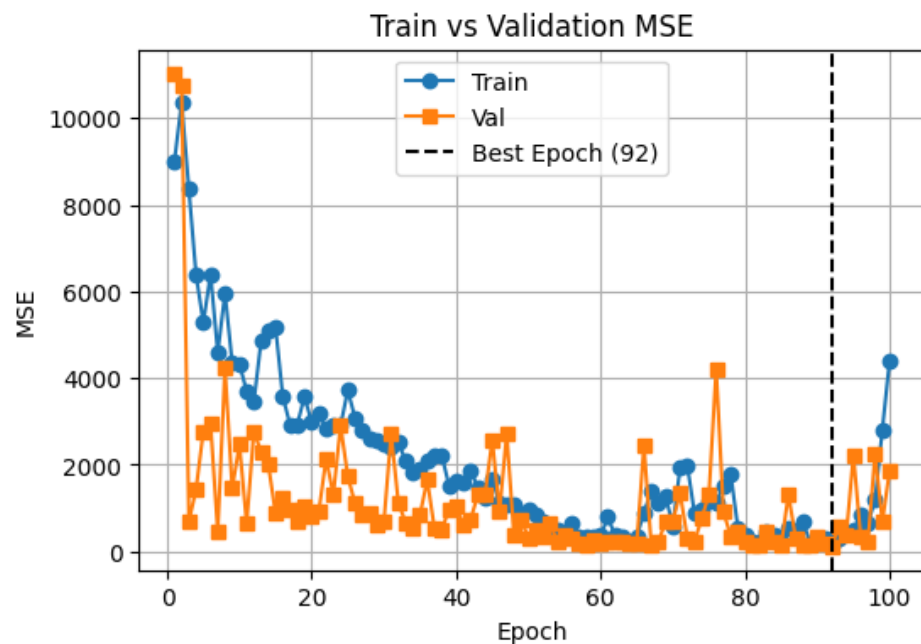


Supervised Cross-Modal Fusion

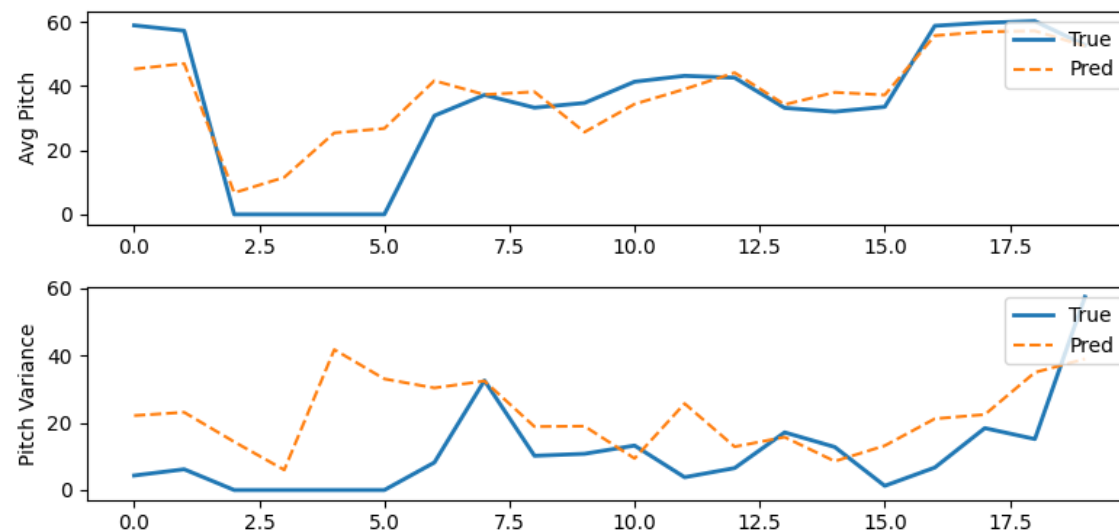


Brain Activity Captures Musical Features

- Trained supervised model for one user in regression task
- Trained with early stopping condition, saving the best model

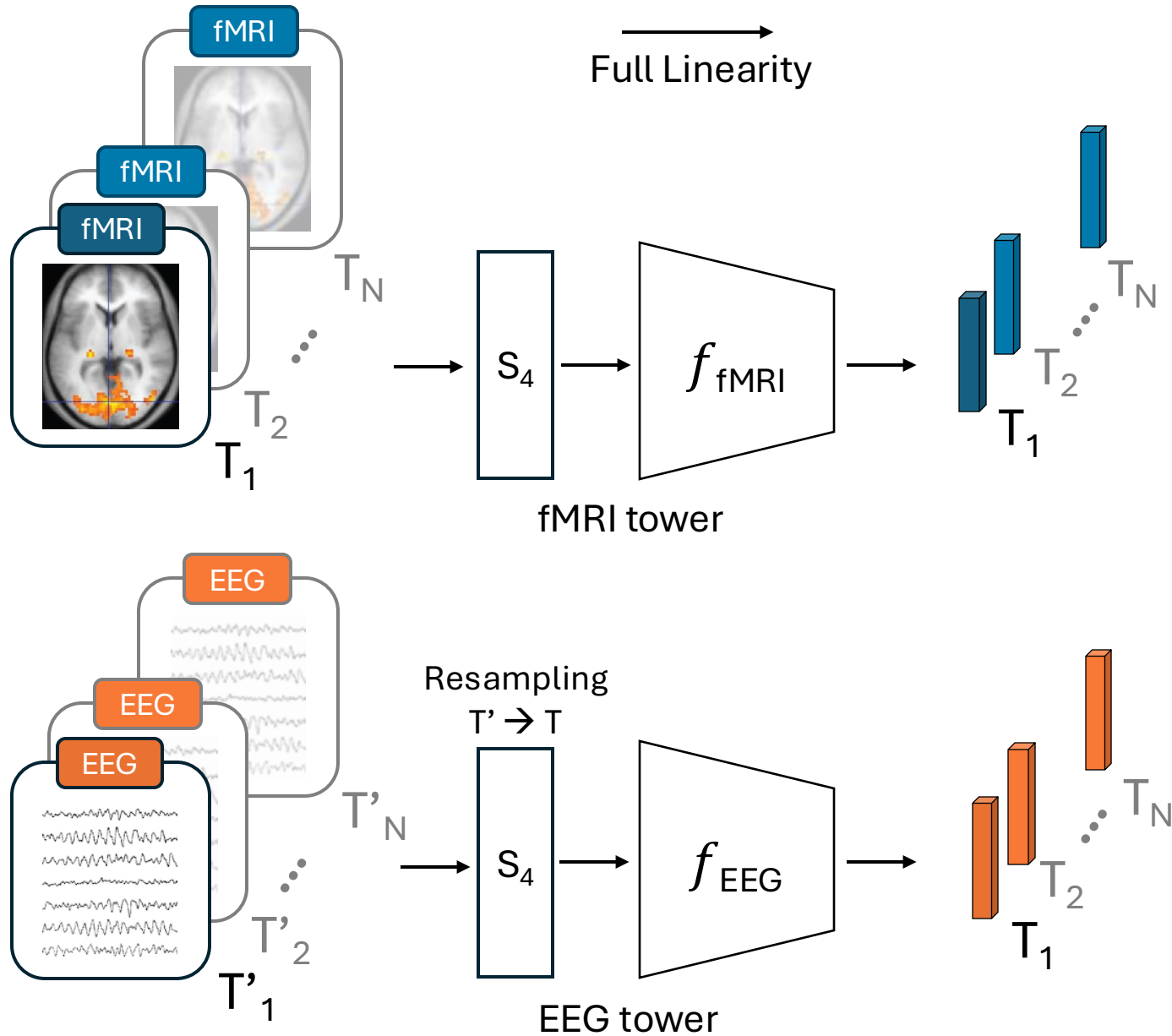


Validation Data Prediction with best model

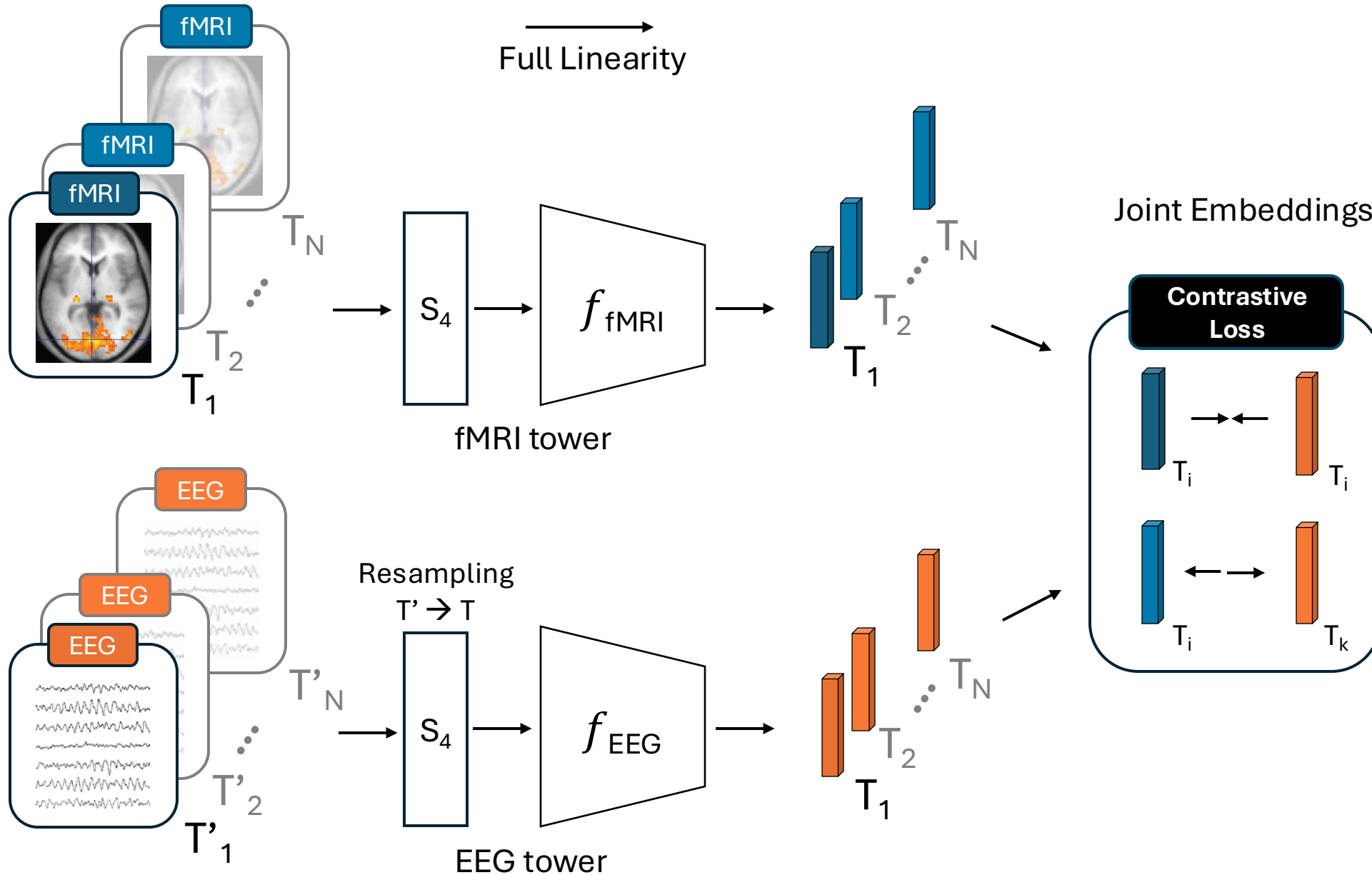


Moving Beyond Task-Specific Embeddings

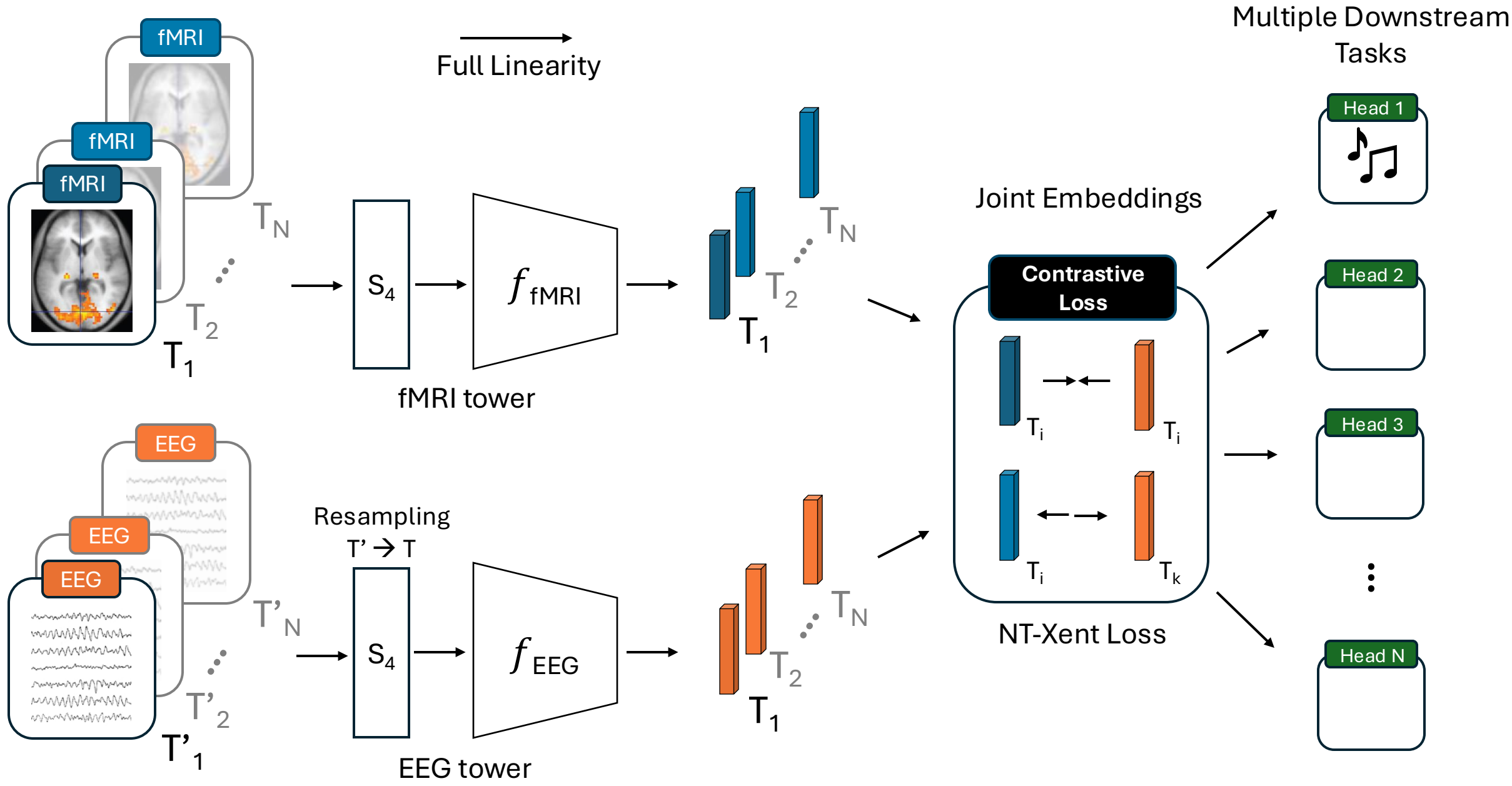
Self-Supervised Cross-Modal Fusion



Self-Supervised Cross-Modal Fusion

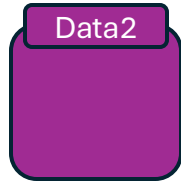
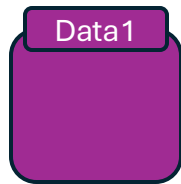


Self-Supervised Cross-Modal Fusion



Self-Supervised Cross-Modal Fusion

Integration of Multiple
Datasets



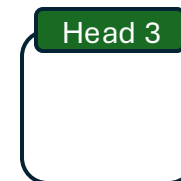
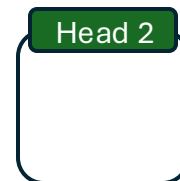
⋮



Joint Embedding Space



Multiple Downstream
Tasks



...



Novelty

- Full linearity of modelling allows for interpretable features
- Modality-agnostic fusion framework

Impact

- Facilitates fusion of modalities with arbitrary sampling rates
- Unsupervised training enables integration of multiple datasets

Rigour

- Noise robustness due to modality-specific signal processing
- Train/Test splitting of data

Refinement

- Demonstrated feasibility of framework on music decoding --> plug and play system on other modalities and tasks
- Documented + modular codebase

Team Yellow



FRANCESCO CENCIARELLI



JAKOB TRÄUBLE



ROHAN KRAJESKI



PABLO SOËTARD



PASCAL WODTKE